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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Examinations March-2021

Common to EC / TE / EE / EI

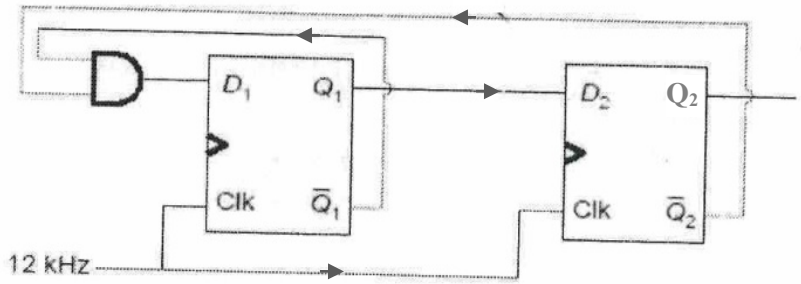
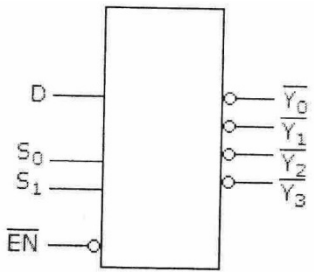
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS

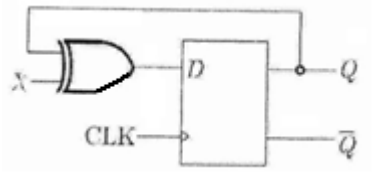
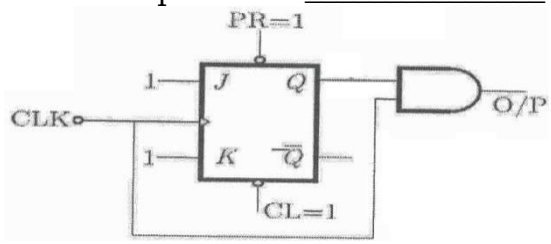
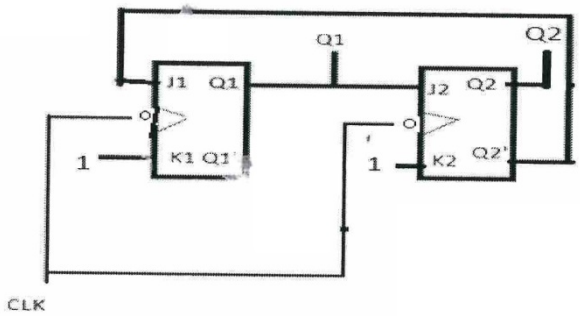
Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	_____ current flows into the output terminal, when the outputs of gate 1 goes low in a logic circuit.	01
	1.2	In the circuit shown in Fig 1.2, frequency of the clock signal is $12kHz$. The frequency of the signal at $Q2$ is _____kHz.  Fig 1.2	02
	1.3	The family of logic devices designed with negative voltage operation, high speed and high power consumption applications is called _____.	01
	1.4	In a processor with register list, the sequence of micro-operations (op-code) for the operations $R_1 \leftarrow R_1 + R_2$.	02
	1.5	A bus organized processor consists of 16 registers with 32 $bits$ in each. The number of selection lines in each multiplexer is _____ and _____ number of 3×8 destination decoders are used.	02
	1.6	For the Demux shown in Fig 1.6, assume the D input is LOW, both S inputs are LOW and the $\overline{EN}$ input is LOW. The status of the $\overline{Y}_0$ outputs is _____.  Fig 1.6	01

1.7	The digital circuit shown in Fig 1.7 works as _____ flip-flop.  Fig 1.7	01
1.8	A $4-bit$ ripple counter consists of flip-flops, where each has a propagation delay of $15ns$. For the counter to recycle form 1111 to 0000, it takes a total time of _____.	02
1.9	The sequences at the output in Fig 1.9, when the flip-flop is initially cleared and clocked for $6-$ pulses is _____.  Fig 1.9	02
1.10	The number of circular right shift operations required to divide the value 10110100 by 4 using shift register is _____.	01
1.11	For the circuit shown in Fig 1.11, the counter is initialized to 00. The sequence of count values at $(Q2, Q1)$ are _____, _____, _____.  Fig 1.11	02
1.12	The initial state of $8-bit$ shift register is 11110000. The data 10110111 needs to be serially loaded (LSB first) into the register. After four clock pulses, the register contains _____.	02
1.13	The status register condition after the operation of unsigned subtraction $(A-B)$, if $A>B$ is _____.	01

PART-B

2	a	Explain with a neat diagram a TTL Inverter with totem-pole output.	06
	b	Assume that a room has three doors and a switch by each door to control a single light in the room. Let X, Y and Z denote the state of all the switches. Assume the light is off if all the switches are open. Closing any switch turns the light ON, but closing another switch will turn the light OFF. Light is ON if all three switches are closed. Construct a truth table and implement the logic using logic gates.	06
	c	Write a five variable K –map for the following function: $f(v,w,x,y,z)=\sum m(4,5,9,11,12,14,15,27,30)+dc(1,17,25,26,31)$	04

3	a	For the given Boolean function $$f(A,B,C,D)=\sum m(2,5,7,8,9,10,13,15)$$ Realize the function using only $4:1$ and $2:1$ multiplexers.	04
	b	A $4\,bit$ ripple carry adder (RCA) realized using full adders. The propagation delay of the XOR, AND, OR gates are $20ns, 15ns, 10ns$ respectively. Assume all the inputs to the $4\,bit$ adder are initially reset to zero. At $T=0$, inputs $X=1100, Y=0100$ & $Z_0=1000$ (carry input at each adder). Calculate the propagation delay.	04
	c	Describe the functioning of a BCD adder with neat logic diagram.	08
		OR	
4	a	Design a $4-$line to $2-$line priority encoder. Include an output E to indicate that at least one input is a one.	05
	b	Implement the following Combinational circuits. i. 2×2 binary multiplier using parallel adder ii. A 3×8 decoder using 2×4 decoder.	05
	c	Prove that a $4-bit$ parallel carry lookahead adder is faster than a $4-bit$ parallel ripple carry adder.	06
5	a	With the help of a suitable example, explain the following operations in a shift register. i. SISO ii. PISO iii. Twisted ring counter.	04
	b	Draw the circuit diagram and explain the operation of a clocked RS flip flop. Derive the characteristic equation.	06
	c	Design and implement a $MOD-7$ synchronous up counter using T flip-flops.	06
		OR	
6	a	Draw the $4-bit$ universal shift register with the help of neat diagram for the following conditions.	07

S0	***S1***	**Operation**
0	0	Buffer
0	1	Shift Right
1	0	Shift Left
1	1	Parallel Output

	b	Convert SR flip-flop to JK flip-flop with the help of function table and excitation table.	05
	c	Design and realize $MOD-4$ asynchronous down counter design using edge triggered JK flip-flop.	04

7 a

It is required to develop a logic circuit for digital quiz that works on a clock and an input from a manual button. Manual switch has to transmit only one HIGH pulse to the circuit. Design a circuit that transits a HIGH pulse with the duration of one clock cycle, when the manual button is presses. The circuit should not transmit next pulse until the button is depressed. Design the logic circuit for the same using finite state machine theory.
(Hint: Use the state diagram given in Fig 7a)

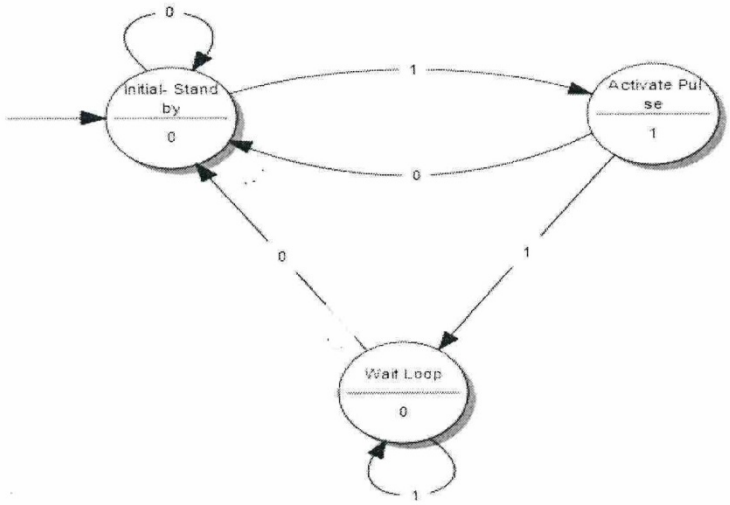


Fig 7a

08

b

Obtain the state diagram from the following logic circuit shown in the Fig 7b of finite state machine.

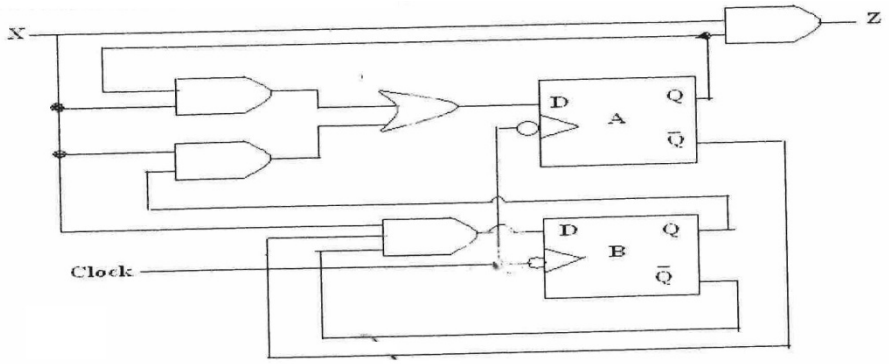


Fig 7b

08

8 a

Explain the processor unit with a 2 –port memory.

04

b

Design an arithmetic circuit with two selection variables, $S1$ and $S2$ that generates the following arithmetic operations. Draw the logic diagram of one typical stage.

$S0$	$S1$	$C_{in} = 0$	$C_{in} = 1$
0	0	$F = A$	$F = A + 1$
0	1	$F = A - B - 1$	$F = A - B$
1	0	$F = B - A - 1$	$F = B - A$
1	1	$F = A + B$	$F = A + B + 1$

06

c

Explain status register and control word register in the processor logic design.

06